

Hybrid Aerial Land Drone

A versatile drone system capable of both aerial navigation and ground-based buggy control with advanced gesture control capabilities using ESP32 microcontrollers.

Project Overview

This project combines aerial and land-based vehicle control systems:

- **Aerial Navigation:** Configured with KK2.1.5 flight controller and FlySky remote control
- **Ground Buggy:** ESP32-controlled vehicle with multiple control methods (Bluetooth, Gesture)
- **Gesture Control:** Custom ESP-NOW wireless communication system using MPU6050 sensor

Hardware Components

Aerial System

- **Flight Controller:** KK2.1.5
- **Remote Control:** FlySky transmitter/receiver
- **Motors:** Quadcopter brushless motors
- **ESCs:** Electronic Speed Controllers
- **Battery:** LiPo battery pack
- **Frame:** Quadcopter frame

Ground Buggy System

- **Microcontroller:** ESP32 Development Board
- **Motor Driver:** L298N Motor Driver Module
- **Motors:** 4x DC Geared Motors
- **Communication:** HC-05 Bluetooth Module
- **Power:** Battery pack for motors and ESP32

Gesture Control System

- **Transmitter:** ESP32 + MPU6050 6-axis gyroscope/accelerometer
- **Receiver:** ESP32 (integrated with buggy control)
- **Communication Protocol:** ESP-NOW (low-latency wireless)
- **Custom PCB:** LED indicators and gesture sensor integration

Features

Aerial Navigation

- ➔ Full 6-axis flight control with KK2.1.5
- □ FlySky remote control with multiple channels
- □ Stable hovering and maneuvering capabilities
- ⚡ Real-time flight adjustments

Ground Buggy Control

- □ 4-wheel drive with individual motor control
- □ Bluetooth smartphone app control
- □ Gesture-based control using hand movements
- □ Multiple movement patterns (forward, backward, turning, strafing)
- □ Real-time response with safety timeout

Gesture Control Features

- □ Hand gesture recognition using MPU6050
- □ ESP-NOW wireless communication (low latency)
- □ Precise motion mapping (yaw, pitch, roll)
- □ Low power consumption
- □ Extended range wireless control

File Structure

```
Hybrid-Aerial-Land-Drone/
├── README.md
├── gesture_control/
│   ├── gesture_transmitter.ino    # ESP32 + MPU6050 transmitter code
│   └── gesture_receiver.ino       # ESP32 receiver + motor control
├── buggy_control/
│   └── buggy_bluetooth.ino        # Bluetooth control for buggy
├── pcb_design/
│   └── (PCB design files - to be added)
└── docs/
    └── (Documentation and schematics)
```

Pin Configurations

Gesture Transmitter (ESP32 + MPU6050)

```
MPU6050 -> ESP32
VCC -> 3.3V
GND -> GND
SDA -> GPIO 21
SCL -> GPIO 22
```

Buggy Control (ESP32 + L298N)

```
L298N -> ESP32
ENA -> GPIO 13
IN1 -> GPIO 12
IN2 -> GPIO 14
IN3 -> GPIO 27
IN4 -> GPIO 26
ENB -> GPIO 25
```

```
HC-05 Bluetooth
RX -> GPIO 22
TX -> GPIO 23
```

Gesture Receiver Motor Control

```
Motor Driver Pins:
- Back Right Motor: IN3(12), IN4(33), ENB(26)
- Back Left Motor: IN1(27), IN2(14), ENA(25)
- Front Right Motor: IN3(12), IN4(33), ENB(26)
- Front Left Motor: IN1(27), IN2(14), ENA(25)
```

Setup Instructions

1. Arduino IDE Setup

```
# Install ESP32 board package in Arduino IDE
# Add the following URL to Additional Board Manager URLs:
https://dl.espressif.com/dl/package_esp32_index.json

# Install required libraries:
- ESP32 by Espressif Systems
- MPU6050 library by Jeff Rowberg
- I2Cdev library
```

2. Gesture Control Setup

1. Transmitter Setup:

- Upload `gesture_transmitter.ino` to ESP32 with MPU6050
- Note the MAC address and update in receiver code
- Calibrate MPU6050 sensor

2. Receiver Setup:

- Upload `gesture_receiver.ino` to buggy ESP32
- Update transmitter MAC address in code
- Test motor connections

3. Bluetooth Control Setup

1. Upload `buggy_bluetooth.ino` to ESP32
2. Pair smartphone with "ESP32_Master"
3. Use compatible Bluetooth terminal app
4. Send commands: 1(forward), 2(backward), 3(left), 4(right), 5(stop)

4. Aerial System Setup

1. KK2.1.5 Configuration:

- Connect ESCs to appropriate ports
- Calibrate ESCs and set failsafe
- Adjust PID settings for stable flight
- Set up aux channels for flight modes

2. FlySky Setup:

- Bind transmitter with receiver
- Configure channel mapping
- Set up failsafe positions
- Test all control surfaces

Control Methods

Gesture Control Commands

- **Forward:** Tilt hand forward (pitch down)
- **Backward:** Tilt hand backward (pitch up)
- **Left:** Tilt hand left (roll left)
- **Right:** Tilt hand right (roll right)

- **Turn Left:** Rotate hand counter-clockwise (yaw left)
- **Turn Right:** Rotate hand clockwise (yaw right)
- **Diagonal movements:** Combine pitch and roll motions

Bluetooth Commands

- Send 1 - Move Forward
- Send 2 - Move Backward
- Send 3 - Turn Left
- Send 4 - Turn Right
- Send 5 - Stop
- Send 6 - Quick Left Turn (400ms)
- Send 7 - Quick Right Turn (400ms)
- Send >20 - Set Speed (21-255)

Technical Specifications

Communication

- **ESP-NOW Frequency:** 2.4GHz
- **Range:** Up to 200 meters (line of sight)
- **Latency:** <10ms
- **Bluetooth:** Classic Bluetooth 2.0/3.0

Power Requirements

- **ESP32:** 3.3V, ~240mA (during transmission)
- **MPU6050:** 3.3V, ~3.9mA
- **Motors:** 6-12V depending on motor specifications
- **Total System:** Recommend 7.4V 2S LiPo battery

Performance

- **Gesture Response Time:** <50ms
- **Motor Control Resolution:** 8-bit PWM (0-255)
- **Movement Accuracy:** ± 2 degrees
- **Operating Temperature:** -10°C to +70°C

Safety Features

- **Signal Timeout:** Motors stop if no signal received for 1 second
- **Emergency Stop:** Multiple stop command options
- **Battery Monitor:** Low voltage warnings (if implemented)
- **Failsafe:** Return to safe position on signal loss

Troubleshooting

Common Issues

1. ESP-NOW Connection Failed

- Verify MAC addresses in both transmitter and receiver
- Ensure both ESP32s are on the same WiFi channel
- Check power supply stability

2. Motors Not Responding

- Verify pin connections and wiring
- Check battery voltage levels
- Test motor driver with multimeter
- Verify PWM signal generation

3. Gesture Control Unresponsive

- Calibrate MPU6050 sensor
- Check I2C connections (SDA/SCL)
- Verify sensor orientation and mounting

4. Bluetooth Connection Issues

- Reset ESP32 and re-pair device
- Check baud rate settings (115200)
- Verify HC-05 module connections

Future Enhancements

- [] Add camera module for FPV
- [] Implement GPS navigation
- [] Add obstacle avoidance sensors
- [] Create mobile app for control
- [] Add telemetry data logging
- [] Implement voice control
- [] Add LED status indicators

- [] Create custom PCB for gesture controller

Contributing

1. Fork the repository
2. Create a feature branch
3. Commit your changes
4. Push to the branch
5. Create a Pull Request

License

This project is open source and available under the [MIT License](#).

Contact

For questions, issues, or contributions, please open an issue in this repository.

⚠ **Safety Warning:** Always operate drones in accordance with local regulations and safety guidelines. Ensure proper battery handling and never leave LiPo batteries unattended while charging.